React JS

Miscellaneous Tasks

Task-1

Create react app and use routing functionality of react to perform the tasks as asked.

Create one file named Main.jsx which contains links for home and about page and routing functionality.

In Home.jsx file add one image and heading(h1) "LJ University"

In About.jsx file add branch list (CSE,IT,CE).

Main.jsx

```
import React from "react";
import { BrowserRouter as Router, Routes, Route, Link } from "react-router-dom";
import Home from "./Home";
import About from "./About";

function Main() {
  return (
    <Router>
      <nav>
        <Link to="/">Home</Link>
        <Link to="/about">About</Link>
      </nav>

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/about" element={<About />} />
      </Routes>
    </Router>
  );
}
export default Main;
```

Home.jsx

```
import React from "react";
import img from "./lj.png"; // image file in src folder

function Home() {
  return (
    <div>
```

```
    <h1>LJ University</h1>
    <img src={img} alt="LJ University" width="300" />
  </div>
 );
}
export default Home;
```

About.jsx

```
import React from "react";

function About() {
 return (
  <div>
    <h2>Branches</h2>
    <ul>
      <li>CSE</li>
      <li>IT</li>
      <li>CE</li>
    </ul>
  </div>
 );
}
export default About;
```

Task 2

Write a ReactJS code in which we have apply filter to skip digits which are less than “10” from the array and display all remaining digits of the array [3,5,11,4,17,8,21,2,26,13,8]. Pass values as props.

Main.jsx

```
import Child from "./Child";

function Main() {
 const arr = [3, 5, 11, 4, 17, 8, 21, 2, 26, 13, 8];

 return (
  <div>
    <h2>Filter Example</h2>
    <Child numbers={arr} />
  </div>
 );}
```

```
export default Main;
```

Child.jsx

```
import React from "react";

function Child(props) {
  const result = props.numbers.filter((num) => num >= 10);

  return (
    <div>
      <h3>Numbers Greater Than or Equal to 10:</h3>
      {result.map((num, index) => (
        <span key={index}>{num} </span>
      ))}
    </div>
  );
}
export default Child;
```

Task 3

Create a React app to perform tasks as asked using functional component:
1)Create a React Router that includes three routes: Home, Food Items, and Contact and implement navigation between these routes.
2)Create a route that displays a Home page, Food Items details page and Contact details page.
3) When a user clicks on Home page it should navigate to the home page and display “Welcome to LJU” in bold in the h1 heading. When a user clicks on a Food Items page, it should navigate to the Food Items Detail page and display three products' information with name, price and description using props.
And when the user clicks on Contact page it should navigate to contact details page and display contact information with blue color font.

Main.jsx

```
import React from "react";
import { BrowserRouter as Router, Routes, Route, Link } from "react-router-dom";
import Home from "./Home";
import FoodItems from "./FoodItems";
import Contact from "./Contact";
```

```
function Main() {
  return (
    <Router>
      <div>
        <Link to="/">Home</Link> |{" "}
        <Link to="/food">Food Items</Link> |{" "}
        <Link to="/contact">Contact</Link>

        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/food" element={<FoodItems />} />
          <Route path="/contact" element={<Contact />} />
        </Routes>
      </div>
    </Router>
  );
}

export default Main;
```

Home.jsx

```
import React from "react";

function Home() {
  return (
    <div>
      <h1><b>Welcome to LJU</b></h1>
    </div>
  );
}

export default Home;
```

Product.js

```
import React from "react";

function Product(props) {
  return (
    <div>
      <h3>{props.name}</h3>
      <p>Price: ₹{props.price}</p>
```

```
    <p>Description: {props.desc}</p>
    <hr />
   </div>
  );
}

export default Product;
```

Fooditems.jsx

```
import React from "react";
import Product from "./Product";

function FoodItems() {
  return (
   <div>
    <h2>Food Items Details</h2>
    <Product     name="Pizza"    price="250" desc="Cheesy Veg Pizza" />
    <Product   name="Burger"     price="120" desc="Spicy Veg Burger" />
    <Product   name="Sandwich"   price="80"   desc="Grilled Vegetable Sandwich"     />
   </div>
  );
}

export default FoodItems;
```

Contact.jsx

```
import React from "react";

function Contact() {
  return (
   <div style={{ color: "blue" }}>
    <h2>Contact Details</h2>
    <p>Email: info@lju.edu.in</p>
    <p>Phone: 9876543210</p>
    <p>Address: Ahmedabad</p>
   </div>
  );
}

export default Contact;
```

Task 4

Create a React function component named File1 that receives two props, productName and price. The component should display the product name and price. Additionally, add a button labeled "Add to cart" to the component. When the button is clicked onClick event triggered and an alert should be displayed with the message "Your Product is Added to cart!".

File1.jsx

```
import React from "react";

function File1(props) {
  const addToCart = () => {
    alert("Your Product is Added to cart!");
  };

  return (
    <div>
      <h2>Product Name: {props.productName}</h2>
      <h3>Price: ₹{props.price}</h3>

      <button onClick={addToCart}>
        Add to cart
      </button>
    </div>
  );
}

export default File1;
```

Main.jsx

```
import React from "react";
import File1 from "./File1";

function Main() {
  return (
    <div>
      <File1   productName="Laptop"  price="50000" />
    </div>
  );
}

export default Main;
```

Task 5

Create a React app:
• Parent component stores language value (“English” or “Hindi”)
• Pass language using useContext
• Child component displays “Welcome” in selected language
• Add button in parent to toggle language

Parent.jsx

```
import React, { createContext, useState } from "react";
import Child from "./Child";

export const LanguageContext = createContext();

function Parent() {
  const [language, setLanguage] = useState("English");

  const toggleLanguage = () => {
    setLanguage(language === "English" ? "Hindi" : "English");
  };

  return (
    <LanguageContext.Provider value={language}>
      <button onClick={toggleLanguage}>
        Toggle Language
      </button>

      <Child />
    </LanguageContext.Provider>
  );
}

export default Parent;
```

Child.jsx

```
import React, { useContext } from "react";
import { LanguageContext } from "./Parent";

function Child() {
  const language = useContext(LanguageContext);

  return (
```

```
  <h2>
    {language === "English" ? "Welcome" : "स्वागत है"}
  </h2>
 );
}

export default Child;
```

Task 6

Create a React app:
• Add Start and Stop buttons
• When Start is clicked, counter increments every second
• When Stop is clicked, counter stops
• Use useEffect with cleanup

Counter.jsx

```
import React, { useState, useEffect } from "react";
function Counter() {
  const [count, setCount] = useState(0);
  const [running, setRunning] = useState(false);

  useEffect(() => {
   let interval;
    if (running) {
      interval = setInterval(() => {
        setCount((prev) => prev + 1);
      }, 1000);
    }
    return () => {
      clearInterval(interval); // Cleanup
    };
  }, [running]);

  return (
    <div>
      <h2>Counter: {count}</h2>
      <button onClick={() => setRunning(true)}>  Start  </button>
      <button onClick={() => setRunning(false)}>  Stop </button>
    </div>
  );}
export default Counter;
```

Task 7

Create a React app:
• Add one textarea
• Display live character count below it
• If characters exceed 100, text color becomes red

```
import React, { useState } from "react";

function CharacterCounter() {
  const [text, setText] = useState("");

  return (
    <div>
      <h2>Character Counter</h2>
<textarea   rows="5"  cols="40"  value={text} onChange={(e) => setText(e.target.value)}      >
</textarea>

      <p style={{ color: text.length > 100 ? "red" : "black" }}>
        Character Count: {text.length}
      </p>
    </div>
  );
}

export default CharacterCounter;
```

Task 8

Create a React application that allows users to toggle between Light and Dark themes. Use useState to manage the theme state and useContext to share this state between components. Define the context inside the same file. When the user clicks a button, the theme should toggle, and the background and text colors should change accordingly. Display the current mode as "Light Mode" or "Dark Mode" in the center of the screen.

Main.jsx (Defines Context, provides it)

```
import React, { useState, createContext } from "react";
import Ur1 from "./Ur1";
// Define context inside Main.jsx
export const ThemeContext = createContext();

function Main() {
```

```
  const [darkMode, setDarkMode] = useState(false);
  const toggleTheme = () => setDarkMode((prev) => !prev);
  const value = {darkMode, toggleTheme};
  return (
    <ThemeContext.Provider value={value}>
      <Ur1 />
    </ThemeContext.Provider>
  )}
export default Main;
```

☑ Ur1.jsx (Consumes Context)

```
import React, { useContext } from "react";
import { ThemeContext } from "./Main"; // Import context from Main.js

function Ur1() {
  const { darkMode, toggleTheme } = useContext(ThemeContext);
  const style = {
    backgroundColor: darkMode ? "#222" : "#eee",
    color: darkMode ? "#fff" : "#000",
    height: "100vh",    display: "flex",    justifyContent: "center",    alignItems: "center",
flexDirection: "column"
  };
  return (
    <div style={style}>
      <h2>{darkMode ? "Dark Mode" : "Light Mode"}</h2>
      <button onClick={toggleTheme}>Toggle Theme</button>
    </div> );}
export default Ur1;
```

Task 9: Create a React app with a student registration form that includes fields for User Name, Age, Gender (radio), and City (dropdown), and validates that all fields are filled and Age is less than or equal to 60 before displaying the submitted data in an alert box.

```
import React, { useState } from 'react';
function StudentForm() {
  const [formData, setFormData] = useState({});

  function handleChange (e){
    const name = e.target.name;
    const value = e.target.value;
   setFormData({ ...formData , [name] :value});
```

```
  };

  function handleSubmit (e) {
    e.preventDefault();

    if (formData.name === undefined ||formData.age === undefined || formData.gender ===
undefined || formData.city===undefined) {
      alert('Enter all details');
    }

    else if ( parseInt(formData.age) >= 60) {
      alert('Age must be 60 or below!');
    }
else{
    alert(`Submitted Data:
Name: ${formData.name}
Age: ${formData.age}
Gender: ${formData.gender}
city: ${formData.city}`);
  };}

  return (
    <div>
      <h2>Registration</h2>
      <form onSubmit={handleSubmit}>
        <div>
          <label>Name:</label><br />
          <input type="text" name="name" onChange={handleChange} />
        </div><br />

        <div>
          <label>Age:</label>
          <input type="number" name="age" onChange={handleChange} />
        </div>

        <div>
          <label>Gender:</label>
          <input type="radio" name="gender" value="Male" onChange={handleChange} /> Male
          <input type="radio" name="gender" value="Female" onChange={handleChange} />
Female
        </div>

        <div>
          <label>city:</label>
```

```
        <select name="city" onChange={handleChange}>
          <option value="Select" default selected hidden >--Select--</option>
          <option value="Ahmedabad" >Ahmedabad</option>
          <option value="Surat">Surat</option>
          <option value="Rajkot">Rajkot</option>
        </select>
      </div>

      <button type="submit">Submit</button>
    </form>
  </div>
 );
}

export default StudentForm;
```

Task 10

Create react app which to perform following task using function component:

- **Create one main file name F1.jsx & other 2 component files F2.jsx & F3.jsx.**
- **Main file contains form with following fields:**
  - **First Name (Input type text)**
  - **Last Name (Input type text)**
  - **Message (Textarea)**
  - **City (Dropdown)**
  - **Gender (Radio Button)**
- **Pass values of all fields from F1.jsx file to F3.jsx file. And display all submitted values in alert box using useContext & useState hook.**

No need to write App.jsx file.

F1.jsx

```
import React, { createContext, useState } from "react";
import F2 from "./F2";

export const UserContext = createContext();

function F1() {
  const [data, setData] = useState({ fname: "", lname: "", message: "", city: "", gender: "" });

  return (
    <UserContext.Provider value={data}>
      <div>
```

```
      <h2>User Form</h2>

      First Name:
      <input  type="text" onChange={(e) => setData({ ...data, fname: e.target.value })} />
      <br /><br />

      Last Name:
      <input  type="text" onChange={(e) => setData({ ...data, lname: e.target.value }) } />
      <br /><br />

      Message:
      <textarea onChange={(e) => setData({ ...data, message: e.target.value })}></textarea>
      <br /><br />

      City:
      <select onChange={(e) => setData({ ...data, city: e.target.value }) } >
        <option value="">Select City</option>
        <option value="Ahmedabad">Ahmedabad</option>
        <option value="Surat">Surat</option>
        <option value="Rajkot">Rajkot</option>
      </select>
      <br /><br />

      Gender:
      <input  type="radio" name="gender" value="Male" onChange={(e) =>
          setData({ ...data, gender: e.target.value })
        }/>  Male

      <input  type="radio"  name="gender" value="Female" onChange={(e) =>
          setData({ ...data, gender: e.target.value })
        } /> Female
     <br /><br />
      <F2 />
    </div>
  </UserContext.Provider>
 );
}

export default F1;
```

F2.jsx

```
import React from "react";
import F3 from "./F3";
```

```
function F2() {
  return (
    <div>
      <F3 />
    </div>
  );
}

export default F2;
```

F3.jsx

```
import React, { useContext } from "react";
import { UserContext } from "./F1";

function F3() {
  const data = useContext(UserContext);

  const showData = () => {
    alert(
      "First Name: " + data.fname +
      "\nLast Name: " + data.lname +
      "\nMessage: " + data.message +
      "\nCity: " + data.city +
      "\nGender: " + data.gender
    );
  };

  return (
    <button onClick={showData}>
      Submit
    </button>
  );
}

export default F3;
```

Task-11

Create a ReactJS program using function component having two input fields for num1 and num2 and two buttons for addition and subtraction of the two numbers. Display the respective outputs on same page using useState hook. Also display alert box as an effect on every time the addition button is clicked using useEffect hook.

```
import React, { useState, useEffect } from "react";

function Calculator() {
  const [num1, setNum1] = useState("");
  const [num2, setNum2] = useState("");
  const [result, setResult] = useState(0);
  const [addClicked, setAddClicked] = useState(false);

  const addition = () => {
    setResult(Number(num1) + Number(num2));
    setAddClicked(!addClicked);
  };

  const subtraction = () => {
    setResult(Number(num1) - Number(num2));
  };

  useEffect(() => {
    if (addClicked) {
      alert("Addition button clicked!");
    }
  }, [addClicked]);

  return (
    <div>
      <h2>Calculator</h2>

      <input  type="number"  placeholder="Enter Number 1" value={num1}
        onChange={(e) => setNum1(e.target.value) />
      <br /><br />

      <input type="number"  placeholder="Enter Number 2"   value={num2}
        onChange={(e) => setNum2(e.target.value)}
      />
      <br /><br />

      <button onClick={addition}>Addition</button>
      <button onClick={subtraction}>Subtraction</button>

      <h3>Result: {result}</h3>
    </div>
  );
}
export default Calculator;
```

Task 12

Create a Student Data Entry Form using React. Fields are Student Name (input Field), Gender (Radio Button values are Boy, Girl), Subjects (Dropdown values are English, Gujarati), Comments (textarea). When the user Click on the Submit button , it will display an alert and use useState hook.

StudentForm.jsx

```
import React, { useState } from "react";

function StudentForm() {
  const [name, setName] = useState("");
  const [gender, setGender] = useState("");
  const [subject, setSubject] = useState("");
  const [comments, setComments] = useState("");

  const handleSubmit = () => {
    alert(
      "Student Name: " + name +
      "\nGender: " + gender +
      "\nSubject: " + subject +
      "\nComments: " + comments
    );
  };

  return (
    <div>
      <h2>Student Data Entry Form</h2>

      Student Name:
      <input  type="text" value={name} onChange={(e) => setName(e.target.value)} />
      <br /><br />

      Gender:
      <input  type="radio" name="gender" value="Boy"  onChange={(e) =>
setGender(e.target.value)} /> Boy

      <input  type="radio" name="gender" value="Girl"  onChange={(e) =>
setGender(e.target.value)}
      /> Girl
      <br /><br />

      Subject:
      <select value={subject} onChange={(e) => setSubject(e.target.value)}>
        <option value="">Select Subject</option>
        <option value="English">English</option>
```

```
      <option value="Gujarati">Gujarati</option>
     </select>
     <br /><br />

     Comments:
     <br />
     <textarea rows="4"  cols="30" value={comments} onChange={(e) =>
setComments(e.target.value)}></textarea>
     <br /><br />

     <button onClick={handleSubmit}>
      Submit
     </button>
    </div>
  );
}

export default StudentForm;
```

Task 13
Create react app to craete exam form. Fields are as under.
Name -text,
Email -email
Password -password,
Gender-radio,
Exam date-datepicker,
Exam center-dropdown
Use useState hook to save the state of the form. Also, add validation for the email and password fields. Display submited values.

```
Examform.jsx
import React, { useState } from "react";

function ExamForm() {
  const [formData, setFormData] = useState({name: "", email: "", password: "", gender: "",
    examDate: "", center: "",});

  const [result, setResult] = useState(null);

  const handleChange = (e) => {
    setFormData({ ...formData, [e.target.name]: e.target.value,});
  };
```

```
const handleSubmit = (e) => {
  e.preventDefault();

  if (!formData.email.includes("@")) {
    alert("Invalid Email");
    return;
  }

  if (formData.password.length < 6) {
    alert("Password must be at least 6 characters");
    return;
  }

  setResult(formData);
};

return (
  <div>
    <h2>Exam Registration Form</h2>

    <form onSubmit={handleSubmit}>
      Name:
      <input  type="text" name="name" onChange={handleChange}/>
      <br /><br />

      Email:
      <input type="email" name="email" onChange={handleChange}/>
      <br /><br />

      Password:
      <input type="password"  name="password"   onChange={handleChange} />
      <br /><br />

      Gender:
   <input  type="radio" name="gender" value="Male" onChange={handleChange}/> Male
   <input type="radio" name="gender" value="Female" onChange={handleChange}/> Female
      <br /><br />

      Exam Date:
      <input type="date" name="examDate" onChange={handleChange}/>
      <br /><br />

      Exam Center:
```

```
        <select
          name="center"
          onChange={handleChange}
        >
          <option value="">Select Center</option>
          <option value="Ahmedabad">Ahmedabad</option>
          <option value="Surat">Surat</option>
          <option value="Rajkot">Rajkot</option>
        </select>
        <br /><br />

        <button type="submit">Submit</button>
      </form>

      {result && (
        <div>
          <h3>Submitted Values</h3>
          <p>Name: {result.name}</p>
          <p>Email: {result.email}</p>
          <p>Password: {result.password}</p>
          <p>Gender: {result.gender}</p>
          <p>Exam Date: {result.examDate}</p>
          <p>Exam Center: {result.center}</p>
        </div>
      )}
    </div>
  );
}

export default ExamForm;
```

Task 14

Write a program to build React app to perform the tasks as asked below using useState hook.

- **Add three buttons “Change Text”, “Change Color”, “Hide/Show”.**
- **Add heading “LJ University” in red color(initial) and also add “React Js Hooks” text in h2 tag.**
- **By clicking on “Change text” button text should be changed to “Welcome students” and vice versa.**
- **By clicking on “Change Color” button change color of text to “blue” and vice versa. This color change should be performed while double clicking on the button.**
- **Initially button text should be “Hide”. While clicking on it the button text should be changed to “Show” and text “React Js Hooks” will not be shown.**

```
import React, { useState } from "react";

function App() {
  const [text, setText] = useState("LJ University");
  const [color, setColor] = useState("red");
  const [show, setShow] = useState(true);

  const changeText = () => {
    setText(
      text === "LJ University"
        ? "Welcome students"
        : "LJ University"
    );
  };

  const changeColor = () => {
    setColor(
      color === "red"
        ? "blue"
        : "red"
    );
  };

  const toggleShow = () => {
    setShow(!show);
  };

  return (
    <div>
      <h1 style={{ color: color }}>
        {text}
      </h1>

      {show && <h2>React Js Hooks</h2>}

      <button onClick={changeText}>
        Change Text
      </button>

      <button onDoubleClick={changeColor}>
        Change Color
      </button>

      <button onClick={toggleShow}>
```

```
      {show ? "Hide" : "Show"}
    </button>
  </div>
 );
}

export default App;
```

Task -15 Implement a character counter that displays the number of characters entered in a textarea input.

```
import { useState } from 'react';
function CharCounter() {
const [text, setText] = useState("");
const handleChange = (e) => {
setText(e.target.value);
};
return (
<div>
<h2>Character Counter</h2>
<textarea value={text} onChange={handleChange} />
<p>Characters: {text.length}</p>
</div>
);
}
export default CharCounter;
```

Task: 16 (Both result has to display)**
Create react app which takes user defined inputs number 1 and number 2 and perform addition and subtraction of the numbers. (Use useState hook)

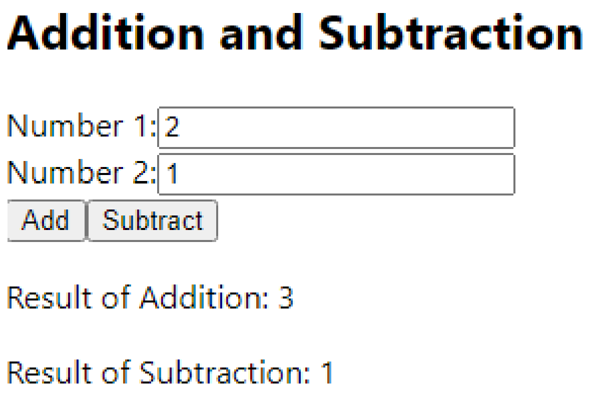
Calc.jsx

```
import React, { useState } from 'react';
function Calc() {
  const [num1, setNum1] = useState(0);
  const [num2, setNum2] = useState(0);
  const [addResult, setAddResult] = useState(0);
  const [subResult, setSubResult] = useState(0);

  const handleAddition = () => {
    const sum = num1 + num2;
    setAddResult(sum);
  };
  const handleSubtraction = () => {
    const difference = num1 - num2;
```

```
  setSubResult(difference);
 };
 return (
  <div>
   <h2>Addition and Subtraction</h2>
   <div>
    <label>
     Number 1:
     <input type="number" value={num1} onChange={(e) =>
setNum1(parseInt(e.target.value))} />
    </label>
   </div>
   <div>
    <label>
     Number 2:
     <input type="number" value={num2} onChange={(e) =>
setNum2(parseInt(e.target.value))} />
    </label>
   </div>
   <div>
    <button onClick={handleAddition}>Add</button>
    <button onClick={handleSubtraction}>Subtract</button>

   </div>
   <div>
    <p>Result of Addition: {addResult}</p>
    <p>Result of Subtraction: {subResult}</p>
   </div>
  </div>
 );
};
export default Calc;
```



Task -17 Create react app which takes user defined inputs number 1 and number 2 and perform addition, subtraction, multiplication, division of the numbers. **(Use useReducer hook)**

```
import React, { useReducer } from "react";
```

```
// Reducer function
function reducer(state, action) {
  const num1 = Number(state.num1);
  const num2 = Number(state.num2);

  switch (action.type) {
    case "SET_NUM1":
      return { ...state, num1: action.value };
    case "SET_NUM2":
      return { ...state, num2: action.value };
    case "ADD":
      return { ...state, result: num1 + num2 };
    case "SUB":
      return { ...state, result: num1 - num2 };
    case "MUL":
      return { ...state, result: num1 * num2 };
    case "DIV":
      return {
        ...state,
        result: num2 !== 0 ? num1 / num2 : "Cannot divide by zero",
      };
    default:
      return state;
  }
}

function Calculator() {
  const [state, dispatch] = useReducer(reducer, { num1: "", num2: "",result: ""});

  return (
    <div style={{ padding: "20px" }}>
      <h2>Simple Calculator (useReducer)</h2>
      <input        type="number"  placeholder="Enter number 1"       value={state.num1}
        onChange={(e) => dispatch({ type: "SET_NUM1", value: e.target.value })}    />

      <input  type="number"  placeholder="Enter number 2"    value={state.num2}
        onChange={(e) => dispatch({ type: "SET_NUM2", value: e.target.value })}   />

      <div style={{ marginTop: "10px" }}>
        <button onClick={() => dispatch({ type: "ADD" })}>Add</button>
        <button onClick={() => dispatch({ type: "SUB" })}>Subtract</button>
        <button onClick={() => dispatch({ type: "MUL" })}>Multiply</button>
        <button onClick={() => dispatch({ type: "DIV" })}>Divide</button>
      </div>
```

```
    <h3>Result: {state.result}</h3>
   </div>
  );
}
export default Calculator;
```

API Random user

```
import React, { useState, useEffect } from 'react';
import axios from 'axios';

const baseURL = "https://randomuser.me/api";

const App = () => {
  const [post, setPost] = useState(null);

  const fetchUser = () => {
    axios.get(baseURL)
      .then((res) => {
        setPost(res.data.results[0]);
      })
      .catch((err) => {
        console.error("Error fetching user:", err);
      });
  };

  useEffect(() => {
    fetchUser(); // Fetch once on initial render
  }, []);

  if (!post) return <p>Loading...</p>;

  const { name, email, picture } = post;

  return (
    <div>
      <h1>Random User</h1>
      <div>
        <img src={picture.large} alt="User" />
        <p>Name: {name.first} {name.last}</p>
        <p>Email: {email}</p>
```

```
    <button onClick={fetchUser}>Load New User</button>
   </div>
  </div>
 );
};

export default App;
```

Advance Level Miscellaneous Tasks [Ref*]

Task 1 : Create react app to filter images based on category while clicking on respective buttons. In example,
Categories – All, Samsung, Vivo and Oneplus.
By clicking on “All” it will display mobiles of all brands. By clicking on specific brand it will display mobiles of respective brand.

```
import React, { useState } from 'react';
import img1 from "./img1.jpg"
import img2 from "./img2.jpg"
import img3 from "./img3.png"
import img4 from "./img4.jpg"
import img5 from "./img5.jpg"
const Gallery = [{ id:1,pic:img1,category:"Samsung"},    { id:2,pic:img2,category:"Mi"},
   { id:3,pic:img3,category:"Oneplus"},    { id:4,pic:img4,category:"Mi"},
   { id:5,pic:img5,category:"Oneplus"},];
function Product () {
   const[images,setImage]=useState(Gallery);
   function handleproduct(Item){
      const finaldata=Gallery.filter((value)=>value.category===Item)
        if(Item !== "All"){ setImage(finaldata); }
      else{ setImage(Gallery) }
   }
  return (
   <div>
      <button onClick={() =>handleproduct('All')}>All</button>
      <button onClick={() =>handleproduct('Samsung')}>Samsung</button>
      <button onClick={() =>handleproduct('Mi')}>Mi</button>
      <button onClick={() =>handleproduct('Oneplus')}>Oneplus</button>
      <div>
         {
            images.map((val)=> {
               return(
               <>
                  <img src={val.pic} height="300" width="300"/>
```

```
                </>
              )
            }) }
          </div>
      </div>
  )}
export default Product
```

Task 2:

Create react app which to perform following task using function component:

• Create one main file name F1.jsx & other 2 component files F2.jsx & F3.jsx.

• Main file contains form with following fields:

o First Name (Input type text)
o Last Name (Input type text)
• Pass values of all fields from F1.jsx file to F3.jsx file. And display all submitted values in alert box using useContext & useState hook. No need to write App.jsx file.

[Also add Message (Textarea), City (Dropdown) , Gender (Radio Button) if required]

```
import { createContext, useState } from 'react';
import F2 from './F2';
const AppContext = createContext();

const AppProvider = () => {
  const [user, setUser] = useState({});
  const [formData, setFormData] = useState({});

  const handleChange = (e) => {
    const { name, value } = e.target;
    setFormData({ ...formData, [name]: value });
  };

  const handleSubmit = (e) => {
    e.preventDefault();
    setUser(formData);
  };
  return (
    <AppContext.Provider value={user}>
      <div>
        <form onSubmit={handleSubmit}>
          <input type="text" name="firstname" onChange={handleChange} />
```

```
            <input type="text" name="lastname" onChange={handleChange} />
            <input type='submit'></input>
          </form>
          <F2 />
        </div>
      </AppContext.Provider>
  );};
export default AppProvider
export { AppContext }
```

F2.jsx

```
import F3 from './F3';
const F2 = () => {
   return (<F3 />)
};
export default F2;
```

F3.jsx

```
import React, { useEffect, useContext } from 'react';
import { AppContext } from './F1';
const UserProfile = () => {
   const user = useContext(AppContext);
useEffect(() => {
         if (Object.keys(user).length > 0) {
        alert(JSON.stringify(user));
      }
   }, [user]);

//Object.keys(user) is a JavaScript method that returns an array of the object's own enumerable property names (i.e., its keys).

   return (
      <div>
         <h2>User Profile</h2>
         <p>Name: {user.firstname} {user.lastname}</p>

      </div>
   );
};
export default UserProfile;
```